

Yousif M. Elsheikh M.

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EDUCATION

Technische Universität Kaiserslautern  2024
Master of Science in Mechanical Engineering and Process Engineering (Spec. Vehicle Engineering) Kaiserslautern, DE
• Thesis: **Reinforcement Learning on Sensor Data and Reaction Grid Maps** at the Robotics Research Lab

University of Khartoum  2016
Bachelor of Science in Mechanical Engineering – Second Class Division One Khartoum, SD

EXPERIENCE

Institute of Aircraft Systems, University of Stuttgart  —  Universität Stuttgart Sep 2024 – present
Research Associate Stuttgart, DE
• Verification and validation of safety-critical cyber-physical systems and deployed neural networks.

IPG Automotive GmbH  Dec 2023 – Aug 2024
Simulation Backend Working Student Karlsruhe, DE
• Development and Implementation of SIL-based V2X communication in CarMaker.
• Achievements – Implemented CPM messages and reception/transmission as well as encoding/decoding with abstract ASN.1.

Adam Riese GmbH  Nov 2022 – Sep 2023
Computer Vision Working Student Stuttgart, DE
• Implementation and delivery of YOLO v7 in a multi-stage object detection pipeline in ONNX runtime.
• Achievements – Trained, validated, benchmarked, and deployed the entire pipeline (live in Aug 2023).

WIS (Würth Industrie Service GmbH & Co. KG)  **WURTH | INDUSTRY** Mar 2022 – Sep 2022
Machine Learning Development Intern Bad Mergentheim, DE
• Development and implementation of 3D computer vision for object localization and recognition.
• Achievements – Successfully implemented 2D detection for computing the Z dimension using point cloud data and mapping to image frame – Confirmed the feasibility of directly learning the point cloud by VoteNet.

Technische Universität Kaiserslautern  Jul 2021 – Mar 2022
Student Research Assistant Kaiserslautern, DE
• I. RRLab (Robotics Research Lab): Underwater object detection using a FLS.
• II. Dept. of Mechanical & Process Engineering: Using GANs and autoencoders to generate GPR sensor data.

Hi-Force Engineering Works Aug 2018 – Feb 2019
Design Engineer Khartoum, SD
• Design conception and standardization using 3D modeling CAD for cost reduction and functionality enhancement.
• Achievements – Prepared complete technical and cost proposals winning a waterworks project (over 100k €).

University of Khartoum  Dec 2016 – Dec 2017
Teaching Assistant Khartoum, SD
• Conducted and managed tutorial sessions. Supervised subjects included: Fluid Mechanics I, Computer Applications, Measurement and Instrumentation, and Hydraulic & Pneumatic Power Transmission.

SELECTED PUBLICATIONS

- **On the Timing Determinism of CUDA and Vulkan SC for Safety-critical AI Workloads on SoC Integrated GPUs.** Yousif M. Elsheikh*, Adrian Raiser*, Fabian Frommherz*, and Zamira Daw. *SAFECOMP 2026*.
- **ASAB: Adversarial Sampling at Code Boundary.** Yassine Akhiat, Yousif M. Elsheikh, Henry Spaet, Zamira Daw, and Bjoern Annighoefer. *IEEE IRAI 2026*.

ADDITIONAL INFORMATION

- **Skills:** Simulation, Machine Learning, Reinforcement Learning, Computer Vision, Deep Learning, Autonomous Driving, Linux, Unreal Engine, CarMaker, Docker, C/C++, Python (Stable Baselines, conda, PyTorch, Pandas, NumPy), Git, CAD (SolidWorks, MATLAB + Simulink) • **Languages:** German (C1/DSH 2; LMU München, TU Kaiserslautern), English (IELTS), Arabic (Native)